

Correction sheet – oral tickets

Algebraic Geometry – Perugia 2026

Oral ticket 1

Question 1

An easy example is $\mathbb{P}_K^1$. It is not affine. One possible argument is that

$$\Gamma(\mathbb{P}_K^1, \mathcal{O}_{\mathbb{P}_K^1}) = K.$$

If $\mathbb{P}_K^1$ were affine, then it would be isomorphic to $\operatorname{Spec} K$, which is impossible, for instance because $\mathbb{P}_K^1$ has more than one point. Equivalently, one may say that $\mathbb{A}_K^2 \setminus \{0\}$ modulo the usual G_m -action gives a projective scheme, not an affine one.

Question 2

The functor

$$F(R) = \{r \in R \mid r^2 = 0\}$$

is representable by

$$\mathbb{Z}[t]/(t^2).$$

Indeed, giving a ring morphism $\mathbb{Z}[t]/(t^2) \rightarrow R$ is the same as choosing an element $r \in R$ such that $r^2 = 0$, namely the image of t .

Oral ticket 2

Question 1

Let $X = \mathbb{P}_A^n = \operatorname{Proj} A[x_0, \dots, x_n]$. The standard affine open subsets $D_+(x_i)$ form a cover of X , and

$$D_+(x_i) \simeq \operatorname{Spec} A \left[\frac{x_0}{x_i}, \dots, \frac{\widehat{x_i}}{x_i}, \dots, \frac{x_n}{x_i} \right].$$

A global regular function on projective space is determined by compatible regular functions on these standard affine charts. One obtains only the constant functions coming from A . Hence

$$\Gamma(\mathbb{P}_A^n, \mathcal{O}_{\mathbb{P}_A^n}) = A.$$

A clean proof is to use the standard fact

$$\Gamma(\operatorname{Proj} S, \mathcal{O}) = S_0$$

for $S = A[x_0, \dots, x_n]$ generated in degree 1 over $S_0 = A$, in this case giving $S_0 = A$.

Question 2

The functor

$$R \longmapsto R^* \times R$$

is representable. The functor $R \mapsto R^*$ is represented by $\mathbb{Z}[u, u^{-1}]$, and $R \mapsto R$ is represented by $\mathbb{Z}[t]$. Therefore the product functor is represented by

$$\mathbb{Z}[u, u^{-1}] \otimes_{\mathbb{Z}} \mathbb{Z}[t] \simeq \mathbb{Z}[u, u^{-1}, t].$$

Oral ticket 3

Question 1

Let

$$A = K[x, y]/(x^2, xy, y^2).$$

The radical of the ideal (x^2, xy, y^2) is (x, y) . Hence $\text{Spec}(A)$ has only one point, corresponding to the maximal ideal (x, y) . Its dimension is 0. The scheme is not reduced, because the classes of x and y are nonzero nilpotent elements.

Question 2

The action of G_m on $\mathbb{A}^1$ by scalar multiplication is, on R -points,

$$R^* \times R \longrightarrow R, \quad (u, t) \longmapsto ut.$$

The corresponding comorphism is

$$\mathbb{Z}[t] \longrightarrow \mathbb{Z}[u, u^{-1}] \otimes_{\mathbb{Z}} \mathbb{Z}[t], \quad t \longmapsto u \otimes t.$$

After identifying the tensor product with $\mathbb{Z}[u, u^{-1}, t]$, this is simply

$$t \longmapsto ut.$$

Oral ticket 4

Question 1

A non-reduced affine scheme with underlying topological space a single point is

$$\text{Spec } K[\varepsilon]/(\varepsilon^2).$$

Its only prime ideal is (ε) , but $\varepsilon \neq 0$ and $\varepsilon^2 = 0$.

A non-reduced affine scheme whose underlying topological space is a line is

$$\text{Spec } K[x, \varepsilon]/(\varepsilon^2).$$

Its reduction is $\text{Spec } K[x]$, but the nilpotent ε gives a non-reduced thickening of the affine line.

Question 2

The translation action of G_a on $\mathbb{A}^1$ is, on R -points,

$$R \times R \longrightarrow R, \quad (a, x) \longmapsto x + a.$$

If s is the coordinate on G_a and t the coordinate on $\mathbb{A}^1$, the corresponding morphism on coordinate rings is

$$\mathbb{Z}[t] \longrightarrow \mathbb{Z}[s] \otimes_{\mathbb{Z}} \mathbb{Z}[t] \simeq \mathbb{Z}[s, t], \quad t \longmapsto t + s.$$

Oral ticket 5

Question 1

The fibre product is computed by tensor product. If the two maps $K[x] \rightarrow K$ send x respectively to 0 and 1, then

$$K \otimes_{K[x]} K \simeq K[x]/(x) \otimes_{K[x]} K[x]/(x-1) \simeq K[x]/(x, x-1) = 0.$$

So the fibre product is empty.

If both maps send x to 0, then

$$K \otimes_{K[x]} K \simeq K[x]/(x) \otimes_{K[x]} K[x]/(x) \simeq K[x]/(x) \simeq K.$$

So the fibre product is $\text{Spec } K$.

Question 2

The functor

$$R \longmapsto \{r \in R \mid r \text{ is nilpotent}\}$$

is not representable by an affine scheme. If it were represented by a ring A , the identity of A would correspond, by Yoneda, to a universal nilpotent element $u \in A$. Since u is nilpotent, we would have $u^N = 0$ for some N .

Now take

$$R = \mathbb{Z}[\varepsilon]/(\varepsilon^{N+1}).$$

The element ε is nilpotent, so it should be the image of u under some morphism $A \rightarrow R$. But then $u^N = 0$ would imply $\varepsilon^N = 0$, which is false in this ring. Contradiction.

Oral ticket 6

Question 1

Consider

$$\mathrm{Spec}(K[t]/(t^2 - a)).$$

Assume first $\mathrm{char} K \neq 2$.

- If $a \neq 0$ is a square in K , then $t^2 - a$ splits with two distinct roots and

$$K[t]/(t^2 - a) \simeq K \times K.$$

The scheme is two reduced K -rational points.

- If a is not a square, then $t^2 - a$ is irreducible and the quotient is a quadratic field extension of K . The scheme has one point with residue field $K(\sqrt{a})$.
- If $a = 0$, then

$$K[t]/(t^2) \simeq K[\varepsilon]/(\varepsilon^2),$$

a non-reduced double point.

If $\mathrm{char} K = 2$, the derivative of $t^2 - a$ is zero. If $a = b^2$, then $t^2 - a = (t - b)^2$, hence the quotient is non-reduced. If a is not a square, the quotient is a purely inseparable quadratic field extension of K .

Question 2

The functor

$$F(R) = \{e \in R \mid e^2 = e\}$$

is represented by

$$\mathbb{Z}[t]/(t^2 - t).$$

Indeed, a ring morphism $\mathbb{Z}[t]/(t^2 - t) \rightarrow R$ is the same as an element $e \in R$ satisfying $e^2 = e$.

Since $t^2 - t = t(t - 1)$ and (t) and $(t - 1)$ are comaximal, the Chinese remainder theorem gives

$$\mathbb{Z}[t]/(t^2 - t) \simeq \mathbb{Z} \times \mathbb{Z}.$$

Geometrically, this is the disjoint union of two copies of $\mathrm{Spec} \mathbb{Z}$, corresponding to the two idempotents 0 and 1.

Oral ticket 7

Question 1

Let $A = K[x, y]$ and

$$U = D(x) \cup D(y) \subset \operatorname{Spec} A = \mathbb{A}_K^2.$$

The complement is

$$V(x) \cap V(y) = V(x, y),$$

namely the origin. Thus

$$U = \mathbb{A}_K^2 \setminus \{0\}.$$

This open subset is not affine. One standard argument is that

$$\Gamma(U, \mathcal{O}_U) = K[x, y],$$

because a regular function on $\mathbb{A}^2 \setminus \{0\}$ extends to $\mathbb{A}^2$. If U were affine, then the canonical morphism $U \rightarrow \operatorname{Spec} \Gamma(U, \mathcal{O}_U) = \mathbb{A}_K^2$ would identify U with $\mathbb{A}_K^2$, which is impossible since the origin is missing.

Question 2

A standard example is

$$\operatorname{Spec}(K[\varepsilon]/(\varepsilon^2)) \longrightarrow \operatorname{Spec} K,$$

induced by the inclusion $K \rightarrow K[\varepsilon]/(\varepsilon^2)$. Both schemes have a single point, so the morphism is bijective on topological spaces. It is not an isomorphism of schemes, because the corresponding ring map is not an isomorphism: the source ring has a nonzero nilpotent, while K is reduced.

Oral ticket 8

Question 1

The morphism

$$\operatorname{Spec}(A_f) \longrightarrow \operatorname{Spec} A$$

is induced by the localization map $A \rightarrow A_f$. Its image is the set of prime ideals $\mathfrak{p} \subset A$ such that $f \notin \mathfrak{p}$, namely

$$D(f) = \operatorname{Spec} A \setminus V(f).$$

Geometrically, one restricts $\operatorname{Spec} A$ to the open subset where the function f does not vanish.

Question 2

The functor

$$G(R) = \{r \in R \mid r^2 = 1\}$$

with multiplication inherited from R^* is μ_2 . It is represented by

$$\mathbb{Z}[t]/(t^2 - 1).$$

The Hopf algebra structure is

$$\Delta(t) = t \otimes t, \quad \varepsilon(t) = 1, \quad \iota(t) = t.$$

These formulas encode multiplication, the identity element, and inversion.

Oral ticket 9

Question 1

Let $R = K[x, y]$ with $\deg(x) = 1$ and $\deg(y) = 2$. Then

$$D_+(x) = \operatorname{Spec} R_{(x)},$$

where $R_{(x)}$ is the degree zero part of $R[x^{-1}]$. Since

$$\deg\left(\frac{y}{x^2}\right) = 0,$$

we get

$$R_{(x)} = K\left[\frac{y}{x^2}\right],$$

and therefore

$$D_+(x) \simeq \mathbb{A}_K^1.$$

Similarly,

$$D_+(y) = \operatorname{Spec} R_{(y)}, \quad R_{(y)} = K\left[\frac{x^2}{y}\right],$$

so

$$D_+(y) \simeq \mathbb{A}_K^1.$$

One should notice, however, that this second chart comes from inverting an element of degree 2, so the usual degree-zero calculation is essential.

Question 2

We compare

$$\mu_2 = \operatorname{Spec}(\mathbb{Z}[t]/(t^2 - 1)) \quad \text{and} \quad \underline{\mathbb{Z}/2\mathbb{Z}} = \operatorname{Spec}(\mathbb{Z} \times \mathbb{Z}).$$

They are not isomorphic over $\mathbb{Z}$. Although

$$t^2 - 1 = (t - 1)(t + 1),$$

the ideals $(t - 1)$ and $(t + 1)$ are not comaximal over $\mathbb{Z}$: their sum contains 2, not 1. Thus $\mathbb{Z}[t]/(t^2 - 1) \not\simeq \mathbb{Z} \times \mathbb{Z}$.

Over a field k with $\operatorname{char} k \neq 2$, the ideals $(t - 1)$ and $(t + 1)$ become comaximal, and

$$k[t]/(t^2 - 1) \simeq k \times k.$$

Thus μ_2 becomes the constant group scheme associated with $\mathbb{Z}/2\mathbb{Z}$. In characteristic 2, however, $t^2 - 1 = (t - 1)^2$, so μ_2 is non-reduced and is not constant.

Oral ticket 10

Question 1

Let $R = K[x, y, z]$, with $\deg(x) = \deg(y) = 1$ and $\deg(z) = 2$. On $D_+(x)$, regular functions are the degree zero elements of $R[x^{-1}]$. We have

$$\deg\left(\frac{y}{x}\right) = 0, \quad \deg\left(\frac{z}{x^2}\right) = 0.$$

Therefore

$$R_{(x)} = K\left[\frac{y}{x}, \frac{z}{x^2}\right].$$

In particular z/x^2 is a regular function on $D_+(x)$, and

$$D_+(x) \simeq \operatorname{Spec} K\left[\frac{y}{x}, \frac{z}{x^2}\right] \simeq \mathbb{A}_K^2.$$

Question 2

The functor

$$R \longmapsto R[x]$$

from rings to sets is not representable. Suppose it were represented by a ring A . By Yoneda, the identity of A would correspond to a universal polynomial

$$P_A(x) \in A[x].$$

This polynomial has finite degree, say d . For any morphism $A \rightarrow R$, its image in $R[x]$ has degree at most d . But the element $x^{d+1} \in \mathbb{Z}[x]$ cannot be obtained in this way. Contradiction.

By contrast, the functor $R \mapsto R$ is representable by $\mathbb{Z}[t]$, since a ring morphism $\mathbb{Z}[t] \rightarrow R$ is the same as choosing the image of t , i.e. an element of R .